

EMAIL THREAT FORENSIC INVESTIGATION REPORT

OFFICIAL DIGITAL FORENSIC EXAMINATION DOSSIER // LAW ENFORCEMENT & SOC RECORD

Case Identifier	Examination Date	Investigation Status	Evidence Hash (SHA-256)
CASE-004	2026-09-05	VERIFIED EXAMINATION	6d3c9e0a5b8f2147e9c2a6d1f4b8...
Classification: PHISHING	Risk Score: 86 / 100	Sample: ITD_Refund_Notice...	File Size: 186 KB

1.0 EXECUTIVE SUMMARY

A forensic examination was conducted on evidence sample 'ITD_Refund_Notice_AY2025.eml' cataloged under reference CASE-004. The electronic message claims transmission from sender 'Income Tax Department' (refunds@incometax-india-gov.in). Automated and deterministic protocol evaluation assessed an authoritative risk score of 86 out of 100, resulting in a forensic threat classification of PHISHING. Authentication protocol verification recorded SPF: FAIL, DKIM: FAIL, and DMARC: FAIL. The message was accompanied by an identified lookalike typosquat target domain (incometaxindia.gov.in), indicating intentional brand impersonation. Routing telemetry analyzed 3 relay hops from origin to terminal destination server.

2.0 EMAIL & EVIDENCE SPECIFICATIONS

Evidence Field	Forensic Data Value	Verification Finding
Original Filename	ITD_Refund_Notice_AY2025.eml	Binary structure parsed and pres...
File Size	186 KB	Standard envelope size
Cryptographic Hash	6d3c9e0a5b8f2147e9c2a6d1f4b8e0c3a7d5f2b9c1e4a8f...	SHA-256 cryptographic integrity ...
Claimed Sender Name	Income Tax Department	Header-extracted display name
Claimed Sender Email	refunds@incometax-india-gov.in	Header From address
Terminal Mail Server	New Delhi, India	Terminal destination host
Originating Node	185.220.101.5	First recorded external IP hop

3.0 AUTHENTICATION PROTOCOL ANALYSIS

Standard email authentication protocols were checked against published Domain Name System (DNS) records. Discrepancies indicate that the sending infrastructure was not authorized to transmit on behalf of the domain.

Protocol	Verdict	Technical Examination Finding
SPF (Sender Policy Framework)	FAIL	Authorized host check: Sending server IP is not listed in domain ...
DKIM (DomainKeys Identified)	FAIL	Cryptographic signature check: Missing or invalid public key cryp...
DMARC (Domain Alignment)	FAIL	Domain alignment policy check: Failed alignment criteria with pub...

4.0 HEADER & ROUTING INFRASTRUCTURE

The message envelope traversed 3 recorded network transit hops. Network infrastructure was cross-referenced with autonomous system records and datacenter threat feeds.

Hop	IP Address	Geographic Location	Relay Timestamp	Forensic Infrastructure Note
#1	185.220.101.5	Amsterdam, Netherlands	2026-09-06 08:12:05	Datacenter flagged
#2	51.89.42.17	London, United Kingdom	2026-09-06 08:12:17	Overseas relay
#3	49.36.80.10	New Delhi, India	2026-09-06 08:12:45	Destination mail server

5.0 LANGUAGE, INTENT & CONTENT ANALYSIS

Content linguistics and social engineering vectors were scored at 24 out of 30 maximum allocated points. Extracted textual phrases were analyzed for deception patterns, urgency cues, and financial solicitation.

Category / Signal	Detected Phrasing / Pattern	Confidence Assessment
FINANCIAL LURE	"click here to claim your pending tax refund of Rs 42,500"	93.0% Match
URGENCY	"unclaimed refunds will be cancelled within 48 hours"	88.0% Match

6.0 REPUTATION & DOMAIN INTELLIGENCE

Attribute	Observed Evidence	Analyst Interpretation
Domain Age	9 days	High anomaly: Domain registered immediately pr...
Typosquatting Target	incometaxindia.gov.in	Deliberate brand impersonation targeting legit...
Reputation Score Contribution	13 / 15 pts	Elevated risk: Associated infrastructure flagg...

7.0 INDICATORS OF COMPROMISE (IOCs)

The following actionable technical indicators were extracted during examination for perimeter blocklisting and SIEM correlation:

Indicator Type	Technical Value	Context / Threat Finding
IP Address (Origin)	185.220.101.5	Originating relay host observed in he...
Email Address	refunds@incometax-india-gov.in	Header From address
Typosquat Target	incometaxindia.gov.in	Target brand impersonated in attack
Forensic Artifact	PAN:ABCDE1234F	Extracted payment/URL artifact
File Hash (SHA-256)	6d3c9e0a5b8f2147e9c2a6d1f4b8e0c3a7d5f2b9c1e4a...	Unique cryptographic hash of message ...

8.0 ATTRIBUTION ASSESSMENT

ATTRIBUTION FINDING: LIKELY SPOOFED DOMAIN IMPERSONATING INCOMETAXINDIA.GOV.IN

Authentication failures combined with a detected lookalike domain registration establish clear domain spoofing.
Note: Attribution conclusions represent technical forensic assessments based on available telemetry and do not constitute definitive legal identification without formal judicial subpoena records.

9.0 CORRELATION & INFRASTRUCTURE RELATIONSHIPS

Graph analysis cross-referenced entities across historical telemetry clusters and previous incident dossiers:

Source Entity	Target Entity	Correlation Basis / Linkage
email1	ip1	same sending IP
email2	ip1	same sending IP
domain1	ip1	same registrar

10.0 RISK ASSESSMENT & SIGNAL WEIGHTING

Overall Risk Score: 86 / 100 (Threshold evaluation: CRITICAL RISK). The score is derived across four standardized examination quadrants:

Quadrant Vector	Allocated Weight	Observed Score	Analytical Contribution
Header & Routing	30 pts	22 pts	Evaluates relay anomaly, ASN hosting, and h...
Authentication Failure	25 pts	27 pts	Evaluates SPF, DKIM signature validity, and...
Language & Intent	30 pts	24 pts	Evaluates urgency, financial diversion, and...
Reputation & History	15 pts	13 pts	Evaluates domain age, typosquatting targets...

11.0 KEY INVESTIGATION FINDINGS

- 1. High-confidence threat signature: Overall threat score exceeds enterprise containment threshold.
- 2. Message failed authentication verification: Sending host lacks authoritative domain alignment.
- 3. Lookalike brand impersonation identified: Attacker targeted legitimate entity 'incometaxindia.gov.in'.
- 4. Originating host flagged: Relay hop #1 hosted on flagged datacenter infrastructure.

12.0 RECOMMENDED INVESTIGATION ACTIONS

- 1. Evidence Preservation: Retain full raw RFC 5322 message headers and message body in forensic storage.
- 2. Network Containment: Implement perimeter firewall block on originating IP 185.220.101.5.
- 3. DNS Mitigation: Issue internal DNS sinkhole rule for identified typosquatting target incometaxindia.gov.in.
- 4. Incident Response: Query mail gateway logs for all inbound messages matching the sending domain within past 30 days.
- 5. Account Safeguards: Enforce credential reset and session termination for any recipient who engaged with message content.

13.0 AUTOMATED NARRATIVE GENERATION ARCHITECTURE NOTE

SYSTEM NOTICE: This document was produced via deterministic evidence extraction from validated email forensic JSON. The

platform architecture is designed to integrate text-generation language models in future backend iterations to draft extended contextual DFIR narratives while maintaining the underlying JSON as the single source of truth.

14.0 EVIDENCE ATTESTATION & LEGAL DISCLAIMER

REPORT IDENTIFIER: REP-CASE-004 | PLATFORM: Email Threat Intelligence & Forensic Platform | EVIDENCE SOURCE: Ingested MIME/MSG Artifact | STATUS: VERIFIED FORENSIC RECORD

LEGAL DISCLAIMER: This document is an analytical examination record compiled from electronic message headers, cryptographic authentication tokens, DNS telemetry, and network routing hops. Findings should be corroborated with original raw headers, provider server logs, and judicial legal process before presentation in legal proceedings.